Quantimite Web

Engineering & Progress Report

Date: November 3, 2026
Branch: `feature/cmp-secure-login`
Project: Next.js App Router Web Platform
STATUS: IN PROGRESS

Executive Summary: This document details the architectural status, recent diagnostic investigations, and code resolutions executed across the **Quantimite Web** platform. Focus areas include resolving WebGL GLSL shader compilation faults in interactive brand assets, analyzing SSR hydration differences induced by client-side browser extensions, and rectifying TypeScript module export discrepancies within the secure CMP internal portal.

1. Key Incidents & Technical Resolutions

ISSUE DESCRIPTION	IMPACTED FILE	STATUS	ROOT CAUSE & SOLUTION
WebGL Shader Compilation Failure 'tex' undeclared, 'texture2D' / 'eyeball' dimension mismatch	`MascotImage.tsx`	RESOLVED	The `eyeball()` shader routine was embedded in vertex shader (`VERT`). Moved into fragment shader (`FRAG`) after uniform sampler. Committed and pushed to GitHub main.
SSR Hydration Attribute Mismatch bis_skin_checked, crxlauncher, bis_register attributes	`layout.tsx`	ANALYZED	Generated by third-party Chrome browser extension (`kiilhncajadbgbmdbdcopdpnmdhlbdle`) mutating the DOM before hydration. Recommended `suppressHydrationWarning` on root HTML tag.
Missing Export in CMP Actions TS2305: 'logoutAdmin' has no exported member	`IdleTimeout.tsx` `actions.ts`	RESOLVED	Server action in `actions.ts` was named `logout` . Updated `IdleTimeout.tsx` to call `logout` , and added export alias `export { logout as logoutAdmin }` to ensure backwards compatibility.

2. Detailed Code Adjustments

A. CMP Portal Logout Resolution

In `src/components/cmp/IdleTimeout.tsx` , the client-side inactivity tracker was updated to reference the authentic server action:

```
// src/components/cmp/IdleTimeout.tsx
import { logout } from "@/app/app-secure-portal-9f8b2c/actions";

export default function IdleTimeout() {
    const timeoutRef = useRef<NodeJS.Timeout | null>(null);
    const INACTIVITY_LIMIT = 5 * 60 * 1000; // 5 minutes

    const handleLogout = async () => {
        await logout();
    };
    ...
}
```

To guard against duplicate naming conventions across submodules, an alias export was appended to `src/app/app-secure-portal-9f8b2c/actions.ts`:

```
// src/app/app-secure-portal-9f8b2c/actions.ts
export async function logout() {
  (await cookies()).delete("cmp_session");
  redirect(`/${SLUG}`);
}

export { logout as logoutAdmin };
```

3. Source Codebase Architecture (`src/`)

Public Marketing & Core App Router

Next.js App Router structure with route groups separating public marketing routes from internal administrative portals:

- **(marketing)/:** Landing pages, service directories, portfolio showcase, careers, contact form, and research blogs.
- **(legal)/:** Compliance and legal documents (Privacy, Terms, Cookies, FAQ).
- **api/:** REST endpoints for chatbot integration, contact submissions, and newsletter subscriptions.

Secure CMP Management Portal

Dedicated internal portal residing under `app-secure-portal-9f8b2c/` protected via signed session cookies:

- **actions.ts:** Server-side login, session creation with HMAC-SHA256, and logout handlers.
- **dashboard/:** Role-based access control (Super Admin, Sales, HR, Content) for inbound CRM leads and job postings.
- **components/cmp/:** Sidebar navigation, activity charts, and inactivity timeouts.

Directory Tree Summary

```
src/
├── app/
│   ├── (legal)/                        # Cookie Policy, FAQ, Privacy, Terms
│   ├── (marketing)/                    # About, Blog, Careers, Contact, Portfolio, Services
│   │   ├── services/                   # Digital Growth, Software Engineering
│   │   ├── layout.tsx & page.tsx       # Main Landing Page Layout
│   ├── api/                            # Careers, Chatbot, CMP Login, Contact endpoints
│   ├── app-secure-portal-9f8b2c/       # Internal CMP Admin Portal
│   │   ├── dashboard/                  # Module routes, layouts, and data views
│   │   ├── actions.ts                  # CMP Server Actions (login, logout, logoutAdmin)
│   │   └── page.tsx                    # Admin Login Screen
│   ├── globals.css & layout.tsx        # Root Layout with Font & Stylesheets
│
├── components/
│   ├── about/, blog/, chat/, common/, forms/
│   ├── home/                           # Hero, Showcase, Process, MascotImage (WebGL)
│   ├── cmp/                            # Sidebar, charts, IdleTimeout
│   ├── layout/                         # Navbar & Footer
│   └── ui/                             # Primitives (Buttons, inputs)
│
├── config/                             # Navigation & Site Metadata
├── lib/                                # CMP Sessions (HMAC-SHA256), DB utilities, helpers
└── types/                              # Application TypeScript Definitions
```

4. Current Git State & Open Work Items

AREA	FILE	BRANCH / STATUS	NOTES
CMP Actions	`src/app/app-secure-portal-9f8b2c/actions.ts`	MODIFIED	Export alias added. Ready to be committed.
Idle Timeout	`src/components/cmp/IdleTimeout.tsx`	UNTRACKED	Inactivity auto-logout component.

Insights / Blog	`src/app/(marketing)/insights/page.tsx`
	`src/components/blog/ArticleHeader.tsx`
	`src/components/blog/RelatedPosts.tsx`
		PENDING	Minor TypeScript check items remaining (null check on BlogPost, missing Lucide imports).